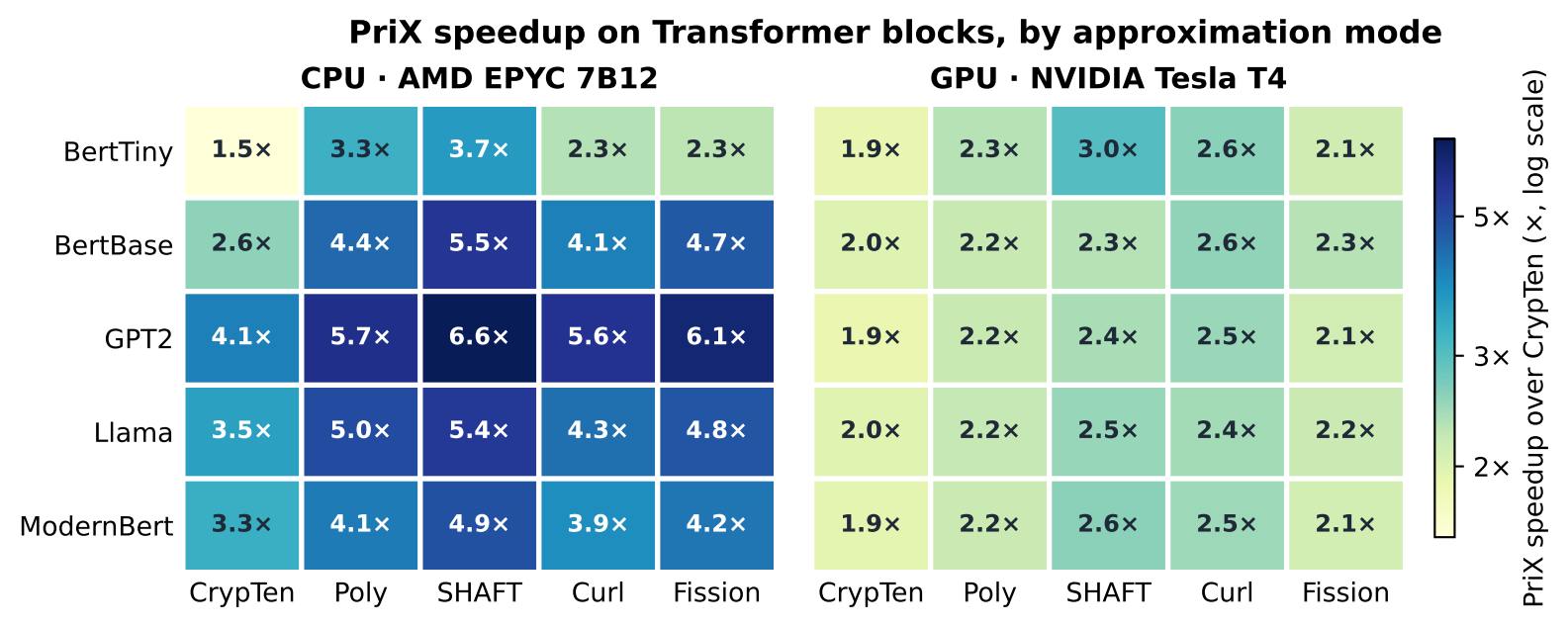
PriX speedup on Transformer blocks, by approximation mode
CPU · AMD EPYC 7B12
GPU · NVIDIA Tesla T4
BertTiny
BertBase
GPT2
Llama
ModernBert
1.5× 3.3× 3.7× 2.3× 2.3×
2.6× 4.4× 5.5× 4.1× 4.7×
4.1× 5.7× 6.6× 5.6× 6.1×
3.5× 5.0× 5.4× 4.3× 4.8×
3.3× 4.1× 4.9× 3.9× 4.2×
1.9× 2.3× 3.0× 2.6× 2.1×
2.0× 2.2× 2.3× 2.6× 2.3×
1.9× 2.2× 2.4× 2.5× 2.1×
2.0× 2.2× 2.5× 2.4× 2.2×
1.9× 2.2× 2.6× 2.5× 2.1×
CrypTen
Poly
SHAFT
Curl
Fission
CrypTen
Poly
SHAFT
Curl
Fission
PriX speedup over CrypTen (×, log scale)
5×
3×
2×